

WAUSAU, WI, USA

WMO: 726463

Lat: 44.929N

Lon: 89.628W

Elev: 1200

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-7.4	-21.1	1.8	-10.9	-16.0	2.4	-6.3	22.1	9.9	19.9	18.1	6.4	300	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	87.6	71.5	84.5	69.5	81.9	67.8	74.4	83.1	72.5	80.8	70.7	78.4	10.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	122.5	79.0	69.8	114.4	77.3	67.9	107.2	75.6	38.6	83.0	36.8	80.8	35.3	78.3	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.4	16.5	-19.0	92.4	5.0	3.8	-22.6	95.1	-25.6	97.3	-28.4	99.5	-32.0	102.2		
(5)			-19.2	77.4	4.9	2.4	-22.8	79.1	-25.7	80.6	-28.4	82.0	-32.0	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	15.4	19.0	30.7	43.7	56.3	66.0	70.2	67.9	60.3	47.4	33.7	21.2		
	DBStd	21.38	12.28	12.20	11.84	9.65	8.46	6.90	5.62	5.48	7.99	8.66	9.74	11.30		
	HDD50	4373	1074	869	610	233	34	0	0	0	10	157	494	892		
	HDD65	7975	1539	1289	1065	640	294	70	17	32	183	550	939	1357		
	CDD50	2343	0	0	10	44	228	480	625	556	320	75	5	0		
	CDD65	469	0	0	0	1	23	99	177	123	43	3	0	0		
	CDH74	3998	0	0	2	26	265	874	1540	918	349	24	0	0		
	CDH80	1029	0	0	0	3	62	236	458	198	71	1	0	0		
	WSAvg	7.6	7.7	7.8	7.8	8.6	8.1	7.3	6.8	6.4	7.0	7.9	8.0	7.6		
	PrecAvg	31.7	1.0	0.9	1.8	2.8	4.1	3.9	3.8	4.2	3.9	2.3	1.9	1.3		
(15) Precipitation	PrecMax	41.7	3.1	2.2	4.6	5.0	7.6	9.4	9.3	9.3	8.7	5.6	4.4	3.0		
	PrecMin	21.4	0.1	0.0	0.2	0.9	1.4	0.5	1.5	0.9	0.6	0.4	0.0	0.4		
	PrecStd	5.6	0.8	0.6	1.3	1.3	1.7	2.1	2.0	1.9	2.1	1.2	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	49.3	69.8	78.2	86.1	90.7	93.3	88.7	86.3	77.5	63.6	50.5		
		MCWB	37.9	42.4	58.2	60.4	67.0	72.5	75.5	72.8	70.9	64.6	53.3	47.0		
	2%	DB	37.8	43.1	60.3	71.0	80.9	85.7	88.3	84.7	81.8	71.8	56.8	42.7		
		MCWB	34.5	38.0	50.7	54.8	63.8	70.3	72.3	70.2	68.2	61.6	49.9	38.3		
	5%	DB	34.5	38.6	52.7	65.4	76.4	82.3	84.8	81.8	78.2	66.6	52.1	38.4		
		MCWB	32.6	34.8	45.4	51.0	60.8	67.4	70.1	68.3	65.8	57.5	45.8	35.5		
	10%	DB	32.2	35.4	46.4	60.5	71.8	79.2	81.9	79.3	74.3	61.8	47.9	35.0		
		MCWB	30.6	32.2	40.2	47.9	58.5	65.7	68.2	66.9	64.0	54.6	42.9	32.8		
	0.4%	WB	38.3	44.6	58.9	62.2	70.7	75.1	78.0	76.3	73.1	68.8	56.0	48.0		
		MCDB	41.9	48.6	67.7	73.5	81.5	85.3	89.1	84.0	81.6	73.8	60.7	50.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.9	38.4	52.7	57.1	66.8	72.7	75.1	73.3	70.5	62.7	51.1	39.3		
		MCDB	37.4	42.5	57.8	68.1	76.1	82.9	84.0	80.7	78.0	68.0	55.7	41.5		
	5%	WB	32.9	35.1	45.9	52.8	63.7	70.4	72.9	71.3	68.4	59.4	46.4	35.5		
		MCDB	34.5	38.4	52.3	62.1	72.5	79.3	81.0	77.9	75.4	65.6	50.9	38.1		
	10%	WB	30.5	32.2	40.6	49.3	60.5	68.2	70.9	69.4	66.0	55.2	43.1	33.0		
		MCDB	32.3	35.0	46.3	59.5	69.1	75.7	78.7	76.1	71.9	61.0	47.6	34.9		
	5% DB	MDBR	14.7	16.2	17.4	19.8	20.9	19.9	20.0	19.1	19.7	17.1	13.9	12.9		
		MCDBR	14.6	17.2	23.4	29.1	26.1	23.3	22.4	21.5	23.7	24.1	20.9	16.1		
		MCWBR	12.7	13.1	15.0	16.0	12.2	10.4	9.5	9.6	11.5	14.5	14.8	13.1		
		MCWBR	13.6	16.0	22.0	25.3	21.8	20.7	20.1	18.3	19.2	21.3	19.3	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.278	0.317	0.346	0.381	0.381	0.374	0.376	0.350	0.315	0.286	0.255		
		taud	2.393	2.300	2.357	2.377	2.320	2.345	2.376	2.372	2.447	2.532	2.523	2.447		
	Ebn at Noon		282	289	291	290	282	281	282	277	277	274	266	267		
		Edh at Noon	23	30	34	36	39	39	37	36	31	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		521	873	1239	1446	1713	1869	1923	1637	1290	810	503	374		
	RadStd		56	81	129	144	136	144	99	108	91	86	52	42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page